1.1 Vectors and Linear Combinations

"You can't add apples and oranges." In a strange way, this is the reason for vectors. We have two separate numbers v_1 and v_2. That pair produces a ***two-dimensional vector*** $\boldsymbol{v}$:

Column vector $\quad \boldsymbol{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ $\quad v_1 =$ first component, $v_2 =$ second component

We write $\boldsymbol{v}$ as a ***column***, not as a row. The main point so far is to have a single letter $\boldsymbol{v}$ (in ***boldface italic***) for this pair of numbers v_1 and v_2 (in *lightface italic*).

Even if we don't add v_1 to v_2, we do *add vectors*. The first components of $\boldsymbol{v}$ and $\boldsymbol{w}$ stay separate from the second components:

VECTOR ADDITION $\quad \boldsymbol{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ and $\boldsymbol{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ add to $\boldsymbol{v} + \boldsymbol{w} = \begin{bmatrix} v_1 + w_1 \\ v_2 + w_2 \end{bmatrix}$.

You see the reason. We want to add apples to apples. Subtraction of vectors follows the same idea: *The components of* $\boldsymbol{v} - \boldsymbol{w}$ *are* $v_1 - w_1$ *and* $v_2 - w_2$.

The other basic operation is *scalar multiplication*. Vectors can be multiplied by 2 or by -1 or by any number c. There are two ways to double a vector. One way is to add $\boldsymbol{v} + \boldsymbol{v}$. The other way (the usual way) is to multiply each component by 2:

SCALAR MULTIPLICATION $\quad 2\boldsymbol{v} = \begin{bmatrix} 2v_1 \\ 2v_2 \end{bmatrix}$ and $-\boldsymbol{v} = \begin{bmatrix} -v_1 \\ -v_2 \end{bmatrix}$.

The components of $c\boldsymbol{v}$ are cv_1 and cv_2. The number c is called a "scalar".

Notice that the sum of $-\boldsymbol{v}$ and $\boldsymbol{v}$ is the zero vector. This is $\mathbf{0}$, which is not the same as the number zero! The vector $\mathbf{0}$ has components 0 and 0. Forgive me for hammering away at the difference between a vector and its components. Linear algebra is built on these operations $\boldsymbol{v} + \boldsymbol{w}$ and $c\boldsymbol{v}$—***adding vectors and multiplying by scalars.***

The order of addition makes no difference: $\boldsymbol{v} + \boldsymbol{w}$ equals $\boldsymbol{w} + \boldsymbol{v}$. Check that by algebra: The first component is $v_1 + w_1$ which equals $w_1 + v_1$. Check also by an example:

$$\boldsymbol{v} + \boldsymbol{w} = \begin{bmatrix} 1 \\ 5 \end{bmatrix} + \begin{bmatrix} 3 \\ 3 \end{bmatrix} = \begin{bmatrix} \mathbf{4} \\ \mathbf{8} \end{bmatrix} \qquad \boldsymbol{w} + \boldsymbol{v} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} + \begin{bmatrix} 1 \\ 5 \end{bmatrix} = \begin{bmatrix} \mathbf{4} \\ \mathbf{8} \end{bmatrix}.$$

Linear Combinations

Combining addition with scalar multiplication, we now form "*linear combinations*" of v and w. Multiply v by c and multiply w by d; then add $cv + dw$.

DEFINITION ***The sum of cv and dw is a linear combination of v and w.***

Four special linear combinations are: sum, difference, zero, and a scalar multiple cv:

$1v + 1w$	=	sum of vectors in Figure 1.1a
$1v - 1w$	=	difference of vectors in Figure 1.1b
$0v + 0w$	=	***zero vector***
$cv + 0w$	=	vector cv in the direction of v

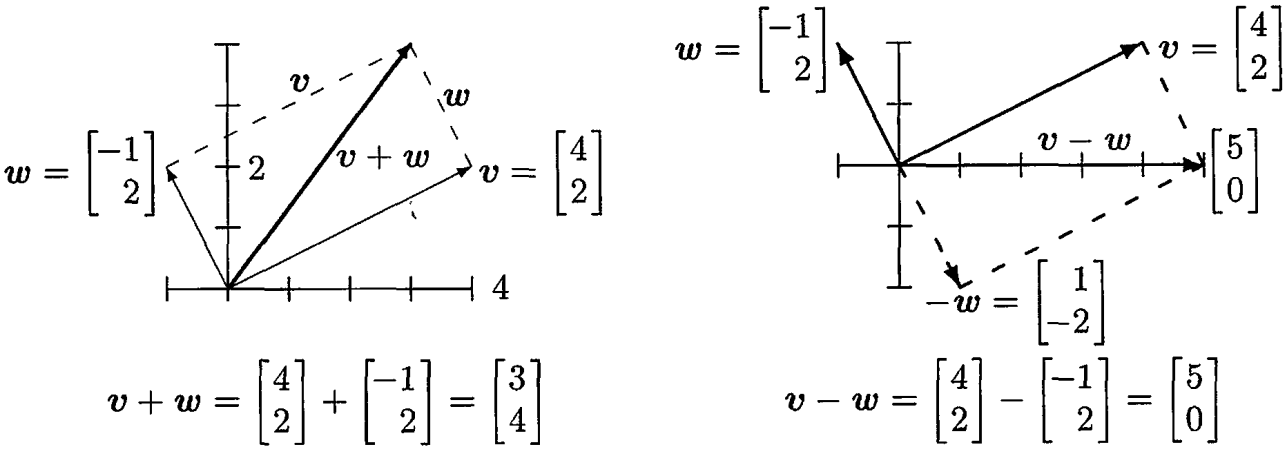
The zero vector is always a possible combination (its coefficients are zero). Every time we see a "space" of vectors, that zero vector will be included. This big view, taking *all* the combinations of v and w, is linear algebra at work.

The figures show how you can visualize vectors. For algebra, we just need the components (like 4 and 2). That vector v is represented by an arrow. The arrow goes $v_1 = 4$ units to the right and $v_2 = 2$ units up. It ends at the point whose x, y coordinates are $4, 2$. This point is another representation of the vector—so we have three ways to describe v:

Represent vector v	Two numbers	Arrow from $(0, 0)$	Point in the plane

We add using the numbers. We visualize $v + w$ using arrows:

Vector addition (head to tail) ***At the end of v, place the start of w.***



$$v + w = \begin{bmatrix} 4 \\ 2 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \qquad v - w = \begin{bmatrix} 4 \\ 2 \end{bmatrix} - \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$$

Figure 1.1: Vector addition $v + w = (3, 4)$ produces the diagonal of a parallelogram. The linear combination on the right is $v - w = (5, 0)$.

We travel along v and then along w. Or we take the diagonal shortcut along $v + w$. We could also go along w and then v. In other words, $w + v$ gives the same answer as $v + w$.

These are different ways along the parallelogram (in this example it is a rectangle). The sum is the diagonal vector $v + w$.

The zero vector $\mathbf{0} = (0, 0)$ is too short to draw a decent arrow, but you know that $v + \mathbf{0} = v$. For $2v$ we double the length of the arrow. We reverse w to get $-w$. This reversing gives the subtraction on the right side of Figure 1.1.

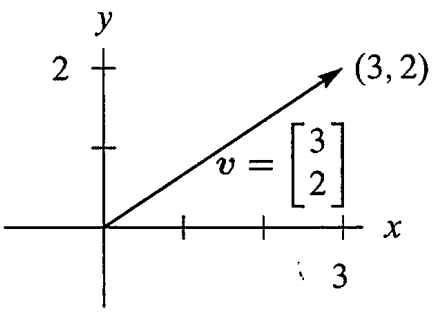
Vectors in Three Dimensions

A vector with two components corresponds to a point in the xy plane. The components of v are the coordinates of the point: $x = v_1$ and $y = v_2$. The arrow ends at this point (v_1, v_2), when it starts from $(0, 0)$. Now we allow vectors to have three components (v_1, v_2, v_3).

The xy plane is replaced by three-dimensional space. Here are typical vectors (still column vectors but with three components):

$$v = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \quad \text{and} \quad w = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \quad \text{and} \quad v + w = \begin{bmatrix} 3 \\ 4 \\ 3 \end{bmatrix}.$$

The vector v corresponds to an arrow in 3-space. Usually the arrow starts at the "origin", where the xyz axes meet and the coordinates are $(0, 0, 0)$. The arrow ends at the point with coordinates v_1, v_2, v_3. There is a perfect match between the ***column vector*** and the ***arrow from the origin*** and the ***point where the arrow ends.***



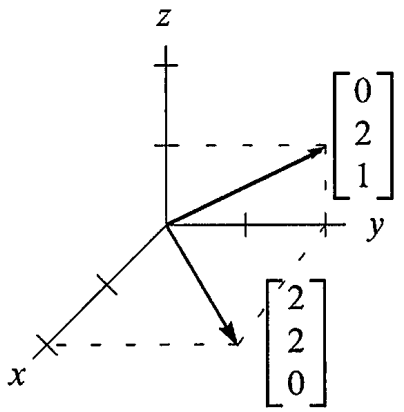


Figure 1.2: Vectors $\begin{bmatrix} x \\ y \end{bmatrix}$ and $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ correspond to points (x, y) and (x, y, z).

From now on $v = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$ ***is also written as*** $v = (1, 1, -1)$.

The reason for the row form (in parentheses) is to save space. But $v = (1, 1, -1)$ is not a row vector! It is in actuality a column vector, just temporarily lying down. The row vector $[1 \quad 1 \quad -1]$ is absolutely different, even though it has the same three components. That row vector is the "transpose" of the column v.

In three dimensions, $v + w$ is still found a component at a time. The sum has components $v_1 + w_1$ and $v_2 + w_2$ and $v_3 + w_3$. You see how to add vectors in 4 or 5 or n dimensions. When w starts at the end of v, the third side is $v + w$. The other way around the parallelogram is $w + v$. Question: Do the four sides all lie in the same plane? *Yes.* And the sum $v + w - v - w$ goes completely around to produce the _____ vector.

A typical linear combination of three vectors in three dimensions is $u + 4v - 2w$:

Linear combination
Multiply by 1, 4, −2
Then add

$$\begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} + 4\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} - 2\begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 9 \end{bmatrix}.$$

The Important Questions

For one vector u, the only linear combinations are the multiples cu. For two vectors, the combinations are $cu + dv$. For three vectors, the combinations are $cu + dv + ew$. Will you take the big step from *one* combination to *all* combinations? Every c and d and e are allowed. Suppose the vectors u, v, w are in three-dimensional space:

1. What is the picture of *all* combinations cu?
2. What is the picture of *all* combinations $cu + dv$?
3. What is the picture of *all* combinations $cu + dv + ew$?

The answers depend on the particular vectors u, v, and w. If they were zero vectors (a very extreme case), then every combination would be zero. If they are typical nonzero vectors (components chosen at random), here are the three answers. This is the key to our subject:

1. The combinations cu fill a ***line***.
2. The combinations $cu + dv$ fill a ***plane***.
3. The combinations $cu + dv + ew$ fill ***three-dimensional space***.

The zero vector $(0, 0, 0)$ is on the line because c can be zero. It is on the plane because c and d can be zero. The line of vectors cu is infinitely long (forward and backward). It is the plane of all $cu + dv$ (combining two vectors in three-dimensional space) that I especially ask you to think about.

Adding all cu on one line to all dv on the other line fills in the plane in Figure 1.3.

When we include a third vector w, the multiples ew give a third line. Suppose that third line is not in the plane of u and v. Then combining all ew with all $cu + dv$ fills up the whole three-dimensional space.

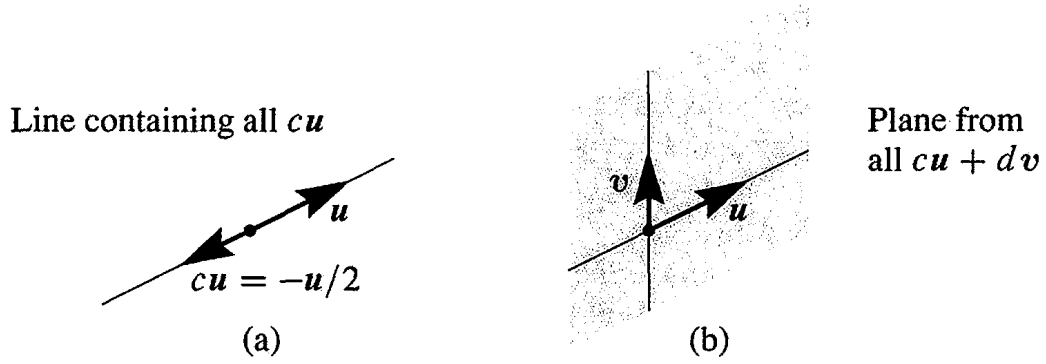


Figure 1.3: (a) Line through u. (b) The plane containing the lines through u and v.

This is the typical situation! **Line**, then **plane**, then **space**. But other possibilities exist. When w happens to be $cu + dv$, the third vector is in the plane of the first two. The combinations of u, v, w will not go outside that uv plane. We do not get the full three-dimensional space. Please think about the special cases in Problem 1.

■ REVIEW OF THE KEY IDEAS ■

1. A vector v in two-dimensional space has two components v_1 and v_2.
2. $v + w = (v_1 + w_1,\ v_2 + w_2)$ and $cv = (cv_1, cv_2)$ are found a component at a time.
3. A linear combination of three vectors u and v and w is $cu + dv + ew$.
4. Take *all* linear combinations of u, or u and v, or u, v, w. In three dimensions, those combinations typically fill a line, then a plane, and the whole space $\mathbf{R}^3$.

■ WORKED EXAMPLES ■

1.1 A The linear combinations of $v = (1, 1, 0)$ and $w = (0, 1, 1)$ fill a plane. *Describe that plane.* Find a vector that is *not* a combination of v and w.

Solution **The combinations $cv + dw$ fill a plane in $\mathbf{R}^3$.** The vectors in that plane allow any c and d. The plane of Figure 1.3 fills in between the "u-line" and the "v-line".

$$\text{Combinations} \quad cv + dw = c\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + d\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} c \\ c + d \\ d \end{bmatrix} \text{ fill a plane.}$$

Four particular vectors in that plane are $(0, 0, 0)$ and $(2, 3, 1)$ and $(5, 7, 2)$ and $(\pi, 2\pi, \pi)$. The second component $c + d$ is always the sum of the first and third components. *The vector* $(1, 2, 3)$ *is not in the plane, because* $2 \neq 1 + 3$.

Another description of this plane through $(0,0,0)$ is to know that $\boldsymbol{n} = (1,-1,1)$ is **perpendicular** to the plane. Section 1.2 will confirm that 90° angle by testing dot products: $\boldsymbol{v} \cdot \boldsymbol{n} = 0$ and $\boldsymbol{w} \cdot \boldsymbol{n} = 0$.

1.1 B For $\boldsymbol{v} = (1,0)$ and $\boldsymbol{w} = (0,1)$, describe all points $c\boldsymbol{v}$ with (**1**) *whole numbers* c (**2**) *nonnegative* $c \geq 0$. Then add all vectors $d\boldsymbol{w}$ and describe all $c\boldsymbol{v} + d\boldsymbol{w}$.

Solution

(**1**) The vectors $c\boldsymbol{v} = (c,0)$ with whole numbers c are **equally spaced points** along the x axis (the direction of $\boldsymbol{v}$). They include $(-2,0), (-1,0), (0,0), (1,0), (2,0)$.

(**2**) The vectors $c\boldsymbol{v}$ with $c \geq 0$ fill a ***half-line***. It is the *positive* x axis. This half-line starts at $(0,0)$ where $c = 0$. It includes $(\pi, 0)$ but not $(-\pi, 0)$.

(**1′**) Adding all vectors $d\boldsymbol{w} = (0,d)$ puts a vertical line through those points $c\boldsymbol{v}$. We have infinitely many ***parallel lines*** from (*whole number* c, *any number* d).

(**2′**) Adding all vectors $d\boldsymbol{w}$ puts a vertical line through every $c\boldsymbol{v}$ on the half-line. Now we have a ***half-plane***. It is the right half of the xy plane (any $x \geq 0$, any height y).

1.1 C Find two equations for the unknowns c and d so that the linear combination $c\boldsymbol{v} + d\boldsymbol{w}$ equals the vector $\boldsymbol{b}$:

$$\boldsymbol{v} = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \qquad \boldsymbol{w} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \qquad \boldsymbol{b} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Solution In applying mathematics, many problems have two parts:

1 *Modeling part* Express the problem by a set of equations.

2 *Computational part* Solve those equations by a fast and accurate algorithm.

Here we are only asked for the first part (the equations). Chapter 2 is devoted to the second part (the algorithm). Our example fits into a fundamental model for linear algebra:

$$\text{Find } c_1, \ldots, c_n \text{ so that } c_1\boldsymbol{v}_1 + \cdots + c_n\boldsymbol{v}_n = \boldsymbol{b}.$$

For $n = 2$ we could find a formula for the $\boldsymbol{c}$'s. The "elimination method" in Chapter 2 succeeds far beyond $n = 100$. For n greater than 1 million, see Chapter 9. Here $n = 2$:

Vector equation
$$c\begin{bmatrix} 2 \\ -1 \end{bmatrix} + d\begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

The required equations for c and d just come from the two components separately:

Two scalar equations
$$\begin{aligned} 2c - d &= 1 \\ -c + 2d &= 0 \end{aligned}$$

You could think of those as two lines that cross at the solution $c = \frac{2}{3}, d = \frac{1}{3}$.

Problem Set 1.1

Problems 1–9 are about addition of vectors and linear combinations.

1 Describe geometrically (line, plane, or all of $\mathbf{R}^3$) all linear combinations of

(a) $\begin{bmatrix}1\\2\\3\end{bmatrix}$ and $\begin{bmatrix}3\\6\\9\end{bmatrix}$ (b) $\begin{bmatrix}1\\0\\0\end{bmatrix}$ and $\begin{bmatrix}0\\2\\3\end{bmatrix}$ (c) $\begin{bmatrix}2\\0\\0\end{bmatrix}$ and $\begin{bmatrix}0\\2\\2\end{bmatrix}$ and $\begin{bmatrix}2\\2\\3\end{bmatrix}$

2 Draw $v=\begin{bmatrix}4\\1\end{bmatrix}$ and $w=\begin{bmatrix}-2\\2\end{bmatrix}$ and $v+w$ and $v-w$ in a single xy plane.

3 If $v+w=\begin{bmatrix}5\\1\end{bmatrix}$ and $v-w=\begin{bmatrix}1\\5\end{bmatrix}$, compute and draw v and w.

4 From $v=\begin{bmatrix}2\\1\end{bmatrix}$ and $w=\begin{bmatrix}1\\2\end{bmatrix}$, find the components of $3v+w$ and $cv+dw$.

5 Compute $u+v+w$ and $2u+2v+w$. How do you know u, v, w lie in a plane?

In a plane $\quad u=\begin{bmatrix}1\\2\\3\end{bmatrix}, \quad v=\begin{bmatrix}-3\\1\\-2\end{bmatrix}, \quad w=\begin{bmatrix}2\\-3\\-1\end{bmatrix}.$

6 Every combination of $v=(1,-2,1)$ and $w=(0,1,-1)$ has components that add to _____. Find c and d so that $cv+dw=(3,3,-6)$.

7 In the xy plane mark all nine of these linear combinations:

$$c\begin{bmatrix}2\\1\end{bmatrix}+d\begin{bmatrix}0\\1\end{bmatrix} \quad \text{with} \quad c=0,1,2 \quad \text{and} \quad d=0,1,2.$$

8 The parallelogram in Figure 1.1 has diagonal $v+w$. What is its other diagonal? What is the sum of the two diagonals? Draw that vector sum.

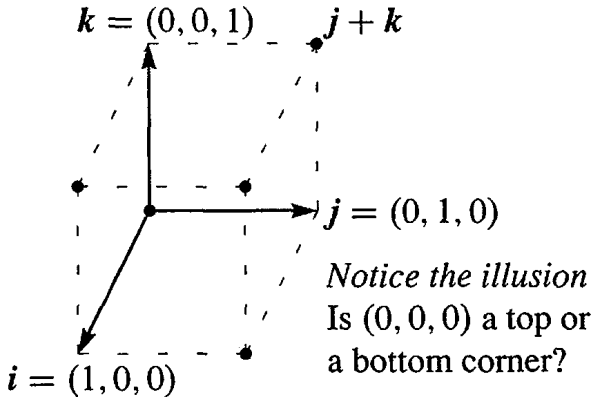
9 If three corners of a parallelogram are $(1,1)$, $(4,2)$, and $(1,3)$, what are all three of the possible fourth corners? Draw two of them.

Problems 10–14 are about special vectors on cubes and clocks in Figure 1.4.

10 Which point of the cube is $i+j$? Which point is the vector sum of $i=(1,0,0)$ and $j=(0,1,0)$ and $k=(0,0,1)$? Describe all points (x,y,z) in the cube.

11 Four corners of the cube are $(0,0,0)$, $(1,0,0)$, $(0,1,0)$, $(0,0,1)$. What are the other four corners? Find the coordinates of the center point of the cube. The center points of the six faces are _____.

12 How many corners does a cube have in 4 dimensions? How many 3D faces? How many edges? A typical corner is $(0,0,1,0)$. A typical edge goes to $(0,1,0,0)$.



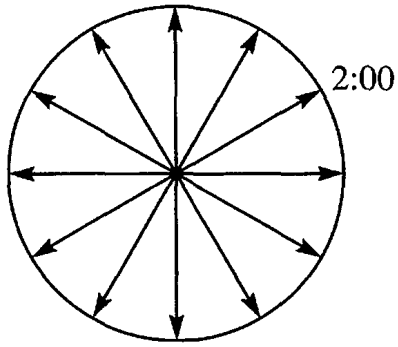


Figure 1.4: Unit cube from i, j, k and twelve clock vectors.

13 (a) What is the sum V of the twelve vectors that go from the center of a clock to the hours 1:00, 2:00, ..., 12:00?

(b) If the 2:00 vector is removed, why do the 11 remaining vectors add to 8:00?

(c) What are the components of that 2:00 vector $v = (\cos\theta, \sin\theta)$?

14 Suppose the twelve vectors start from 6:00 at the bottom instead of $(0, 0)$ at the center. The vector to 12:00 is doubled to $(0, 2)$. Add the new twelve vectors.

Problems 15–19 go further with linear combinations of v and w (Figure 1.5a).

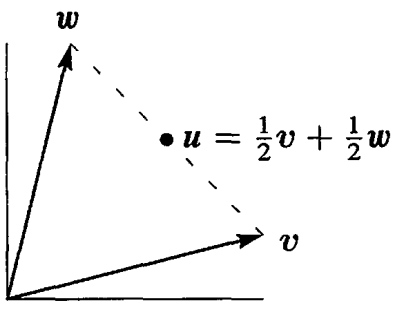
15 Figure 1.5a shows $\frac{1}{2}v + \frac{1}{2}w$. Mark the points $\frac{3}{4}v + \frac{1}{4}w$ and $\frac{1}{4}v + \frac{1}{4}w$ and $v + w$.

16 Mark the point $-v + 2w$ and any other combination $cv + dw$ with $c + d = 1$. Draw the line of all combinations that have $c + d = 1$.

17 Locate $\frac{1}{3}v + \frac{1}{3}w$ and $\frac{2}{3}v + \frac{2}{3}w$. The combinations $cv + cw$ fill out what line?

18 Restricted by $0 \le c \le 1$ and $0 \le d \le 1$, shade in all combinations $cv + dw$.

19 Restricted only by $c \ge 0$ and $d \ge 0$ draw the "cone" of all combinations $cv + dw$.



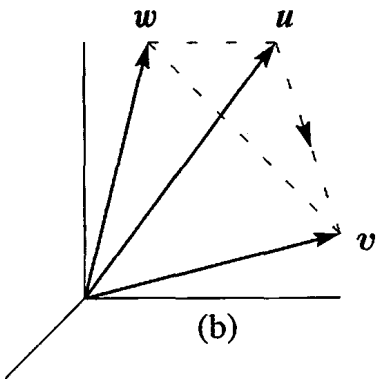


Figure 1.5: Problems **15–19** in a plane Problems **20–25** in 3-dimensional space

Problems 20–25 deal with u, v, w in three-dimensional space (see Figure 1.5b).

20 Locate $\frac{1}{3}u + \frac{1}{3}v + \frac{1}{3}w$ and $\frac{1}{2}u + \frac{1}{2}w$ in Figure 1.5b. Challenge problem: Under what restrictions on c, d, e, will the combinations $cu + dv + ew$ fill in the dashed triangle? To stay in the triangle, one requirement is $c \geq 0, d \geq 0, e \geq 0$.

21 The three sides of the dashed triangle are $v - u$ and $w - v$ and $u - w$. Their sum is _____. Draw the head-to-tail addition around a plane triangle of $(3, 1)$ plus $(-1, 1)$ plus $(-2, -2)$.

22 Shade in the pyramid of combinations $cu + dv + ew$ with $c \geq 0, d \geq 0, e \geq 0$ and $c + d + e \leq 1$. Mark the vector $\frac{1}{2}(u + v + w)$ as inside or outside this pyramid.

23 If you look at *all* combinations of those u, v, and w, is there any vector that can't be produced from $cu + dv + ew$? Different answer if u, v, w are all in _____.

24 Which vectors are combinations of u and v, and *also* combinations of v and w?

25 Draw vectors u, v, w so that their combinations $cu + dv + ew$ fill only a line. Find vectors u, v, w so that their combinations $cu + dv + ew$ fill only a plane.

26 What combination $c\begin{bmatrix}1\\2\end{bmatrix} + d\begin{bmatrix}3\\1\end{bmatrix}$ produces $\begin{bmatrix}14\\8\end{bmatrix}$? Express this question as two equations for the coefficients c and d in the linear combination.

27 *Review Question.* In xyz space, where is the plane of all linear combinations of $i = (1, 0, 0)$ and $i + j = (1, 1, 0)$?

Challenge Problems

28 Find vectors v and w so that $v + w = (4, 5, 6)$ and $v - w = (2, 5, 8)$. This is a question with _____ unknown numbers, and an equal number of equations to find those numbers.

29 Find two different combinations of the three vectors $u = (1, 3)$ and $v = (2, 7)$ and $w = (1, 5)$ that produce $b = (0, 1)$. Slightly delicate question: If I take any three vectors u, v, w in the plane, will there always be two different combinations that produce $b = (0, 1)$?

30 The linear combinations of $v = (a, b)$ and $w = (c, d)$ fill the plane unless _____. Find four vectors u, v, w, z with four components each so that their combinations $cu + dv + ew + fz$ produce all vectors (b_1, b_2, b_3, b_4) in four-dimensional space.

31 Write down three equations for c, d, e so that $cu + dv + ew = b$. Can you somehow find c, d, and e?

$$u = \begin{bmatrix}2\\-1\\0\end{bmatrix} \quad v = \begin{bmatrix}-1\\2\\-1\end{bmatrix} \quad w = \begin{bmatrix}0\\-1\\2\end{bmatrix} \quad b = \begin{bmatrix}1\\0\\0\end{bmatrix}.$$